

NEW CHITOSE AP, Japan

WMO: 474250

Lat: 42.775N

Lon: 141.692E

Elev: 82

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	3.1	-5.9	4.0	0.2	-1.9	5.0	4.7	27.9	31.3	25.6	30.3	3.3	170	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	81.0	71.9	78.7	70.9	75.6	69.3	74.5	78.8	72.7	76.4	71.1	74.5	11.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	124.1	77.5	71.6	116.9	75.1	69.9	110.3	73.4	38.1	79.6	36.4	76.6	35.0	75.7	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.6	20.0	DB	-8.1	86.6	4.6	2.8	-11.4	88.6	-14.1	90.3	-16.7	91.8	-20.1	93.8	(4)
			WB	-8.5	77.1	4.6	2.0	-11.9	78.6	-14.6	79.7	-17.1	80.9	-20.5	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	19.4	20.7	30.0	41.9	52.0	59.3	66.8	69.8	63.0	50.5	38.1	25.3
	DBStd	18.35	6.79	7.08	6.62	5.61	5.10	4.76	4.65	4.58	5.68	6.09	7.19	7.42
	HDD50	3874	948	820	619	250	38	1	0	0	0	70	362	766
	HDD65	7651	1413	1240	1084	693	404	178	35	11	104	451	807	1231
	CDD50	2000	0	0	0	7	99	281	521	612	390	85	5	0
	CDD65	302	0	0	0	0	0	8	91	159	44	0	0	0
	CDH74	1028	0	0	0	0	6	31	278	575	138	0	0	0
	CDH80	147	0	0	0	0	1	3	34	99	10	0	0	0
Wind	WSAvg	9.1	7.5	8.3	9.7	11.2	11.3	10.2	9.7	9.1	8.5	8.5	8.3	7.4
Precipitation	PrecAvg	48.5	3.0	2.5	2.2	2.9	4.3	3.6	5.7	6.5	6.5	4.1	3.9	3.5
	PrecMax	61.8	4.3	5.6	4.0	5.2	7.4	10.0	11.9	13.3	10.9	5.9	8.1	6.6
	PrecMin	39.0	1.8	1.0	0.9	0.3	1.6	1.3	2.7	2.9	1.7	1.7	2.4	2.2
	PrecStd	5.6	0.6	1.0	0.8	1.4	1.5	1.7	2.1	2.7	2.6	1.1	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	41.3	51.6	66.0	73.3	77.4	83.9	86.4	80.9	70.1	60.9	49.9
		MCWB	36.8	37.2	43.1	51.8	57.6	65.2	72.2	76.0	70.9	61.5	54.5	47.3
	2%	DB	37.0	37.8	46.4	59.4	68.3	73.4	80.3	82.6	77.3	66.5	57.3	44.7
		MCWB	34.2	34.6	40.1	48.6	55.9	63.8	71.9	73.3	68.5	59.3	52.2	41.2
	5%	DB	33.9	35.9	42.9	55.6	64.8	70.2	77.1	79.2	75.1	64.4	54.0	40.9
		MCWB	31.3	32.9	38.2	46.6	54.0	62.2	70.0	71.9	67.9	57.8	49.5	37.4
	10%	DB	31.9	33.9	40.7	52.1	62.3	67.8	73.8	77.0	73.1	62.3	51.5	37.2
		MCWB	29.5	31.0	36.5	44.5	53.3	61.0	68.7	70.8	66.9	56.7	47.0	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	38.5	44.3	53.1	60.5	67.8	75.8	77.6	74.4	64.7	56.7	48.1
		MCDB	38.7	39.6	47.9	61.8	68.4	74.2	80.5	84.7	77.6	67.9	59.7	49.7
	2%	WB	34.0	35.5	41.2	50.7	58.2	65.3	73.1	75.4	72.2	61.5	53.7	41.8
		MCDB	35.7	37.2	44.6	56.8	64.9	71.1	77.5	80.4	75.6	65.0	56.8	43.8
	5%	WB	31.8	33.1	38.8	48.6	56.4	63.6	71.4	73.7	70.3	59.5	50.6	37.3
		MCDB	33.4	35.3	41.9	53.8	62.5	68.5	75.2	77.4	73.3	63.2	53.6	39.9
	10%	WB	28.9	31.1	36.6	46.1	54.8	61.9	69.7	72.2	67.9	57.1	47.4	33.8
		MCDB	31.3	33.6	39.9	50.4	60.3	66.3	72.9	75.6	71.6	61.1	50.6	36.2
Mean Daily Temperature Range	5% DB	MDBR	19.3	19.8	15.9	17.4	16.9	13.8	11.5	11.8	16.1	19.1	17.0	17.3
		MCDBR	18.1	18.8	17.2	23.4	22.8	18.6	16.1	15.1	16.9	19.9	21.0	18.9
		MCWBR	16.4	16.3	13.0	14.3	12.4	10.3	8.8	8.1	9.8	13.7	17.1	16.4
	5% WB	MCDBR	17.8	18.2	16.2	20.0	18.5	15.4	12.9	12.5	12.5	17.0	19.4	17.7
		MCWBR	17.0	16.6	13.3	13.5	11.3	9.3	7.9	7.6	8.9	13.2	17.3	16.4
Clear-Sky Solar Irradiance	taub	0.340	0.391	0.465	0.526	0.544	0.525	0.511	0.480	0.420	0.407	0.388	0.355	
	taud	2.044	1.897	1.795	1.786	1.837	1.965	2.072	2.176	2.297	2.195	2.143	2.059	
	Ebn at Noon	239	244	240	237	238	243	245	248	256	240	221	219	
	Edh at Noon	35	48	61	66	65	57	51	44	36	36	32	32	
All-Sky Solar Radiation	RadAvg	488	743	1111	1453	1620	1573	1425	1349	1169	859	518	393	
	RadStd	32	55	78	148	140	130	155	102	105	47	45	33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page